

E-wallet Analysis Report

Droppping Bill Payment Success Rate



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Executive Summary

SITUATION Business Context	- Total transaction volume expanded significantly by 20% between July and August 2023. - Billing Payment is the primary revenue driver, accounting for 37% of total platform volume. - Continuous rollout of 4 app updates (Jul 28 – Aug 22, latest v3.9.9) to rapidly resolve system bugs.
COMPLICATION Business problems	- Overall success rate dropped from 92% to 87% in August, with Billing Payment suffering the sharpest decline (-14%). - System overload during peak hours might impact backend infrastructure and third party payment gateways. - Stricter security protocols (OTP, 3DS/MFA) and fragmented biller integrations might cause high drop-offs across diverse user segments.
QUESTIONS Business concerns	- Did internal system/app updates, backend latency, or external third party biller integration cause the success rate drop? - Which specific step in the payment flow (authentication, input, selection) and device/platform shows the highest error rate? - Which user cohorts (age, digital literacy, active level) are most affected?
ANSWERS Business solutions	Engineering & DevOps: Extend timeout limits for 3DS and API, optimize OTP delivery speed, upgrade backend infrastructure to avoid gateway latency. Product & UX/UI: Redesign payment authentication UI to reduce friction, run A/B tests on updated payment flows Marketing: Set up real-time monitoring dashboards to track conversion bottlenecks, conduct CSAT/NPS feedback analysis on reviews. Customer Service: Implement AI Chatbots and automated smart notifications to assist users; categorize support tickets by error type for quick response.

Agenda

Opening

**Executive
Summary**

Agenda

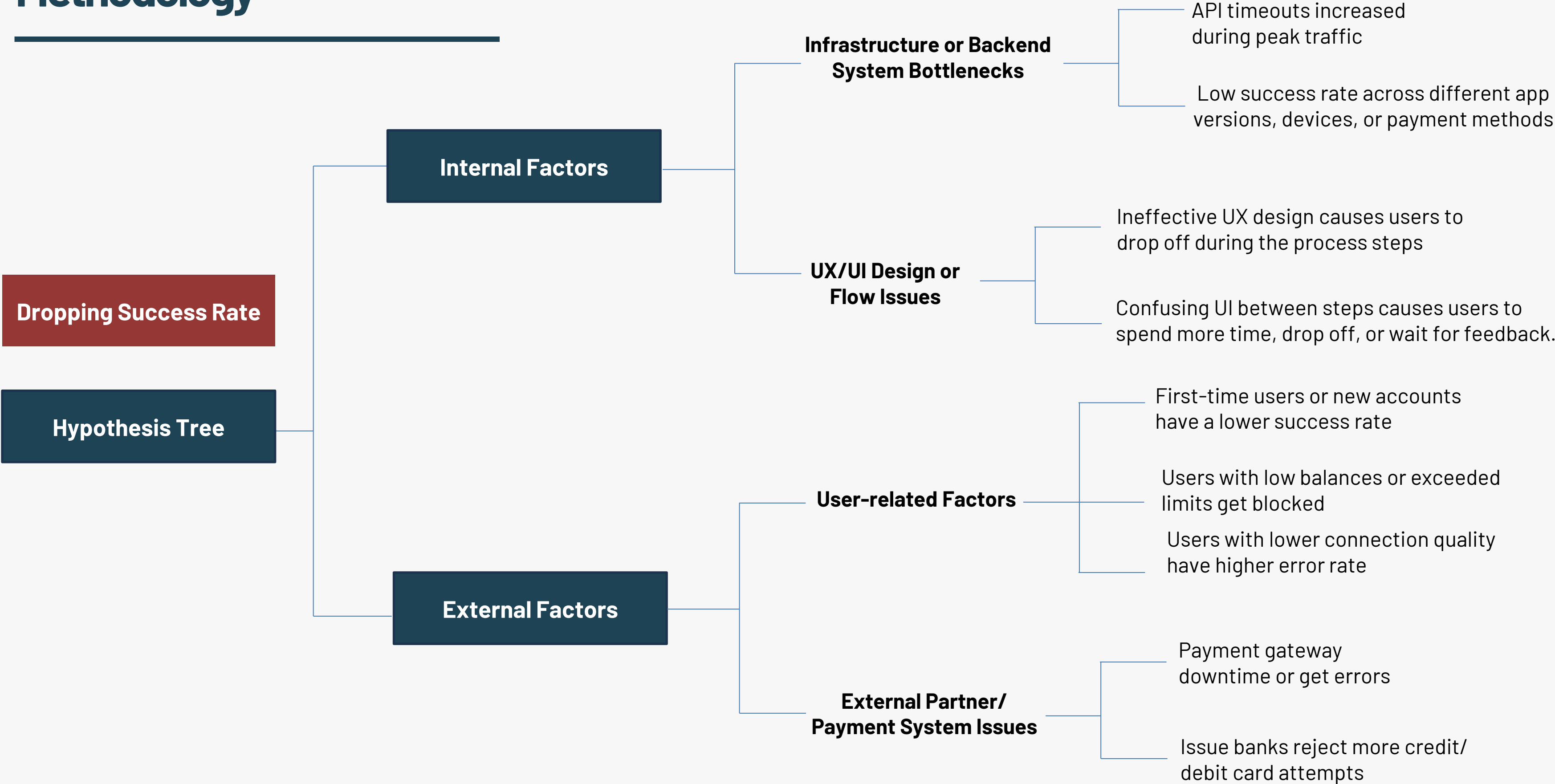
Methodology

Analysis

Findings

Recommendations

Methodology



Using the Issue Tree, we hypothesized possible factors that led to the dropping success rate, including Internal and External factors.

Payment Flow & Involved Stakeholders

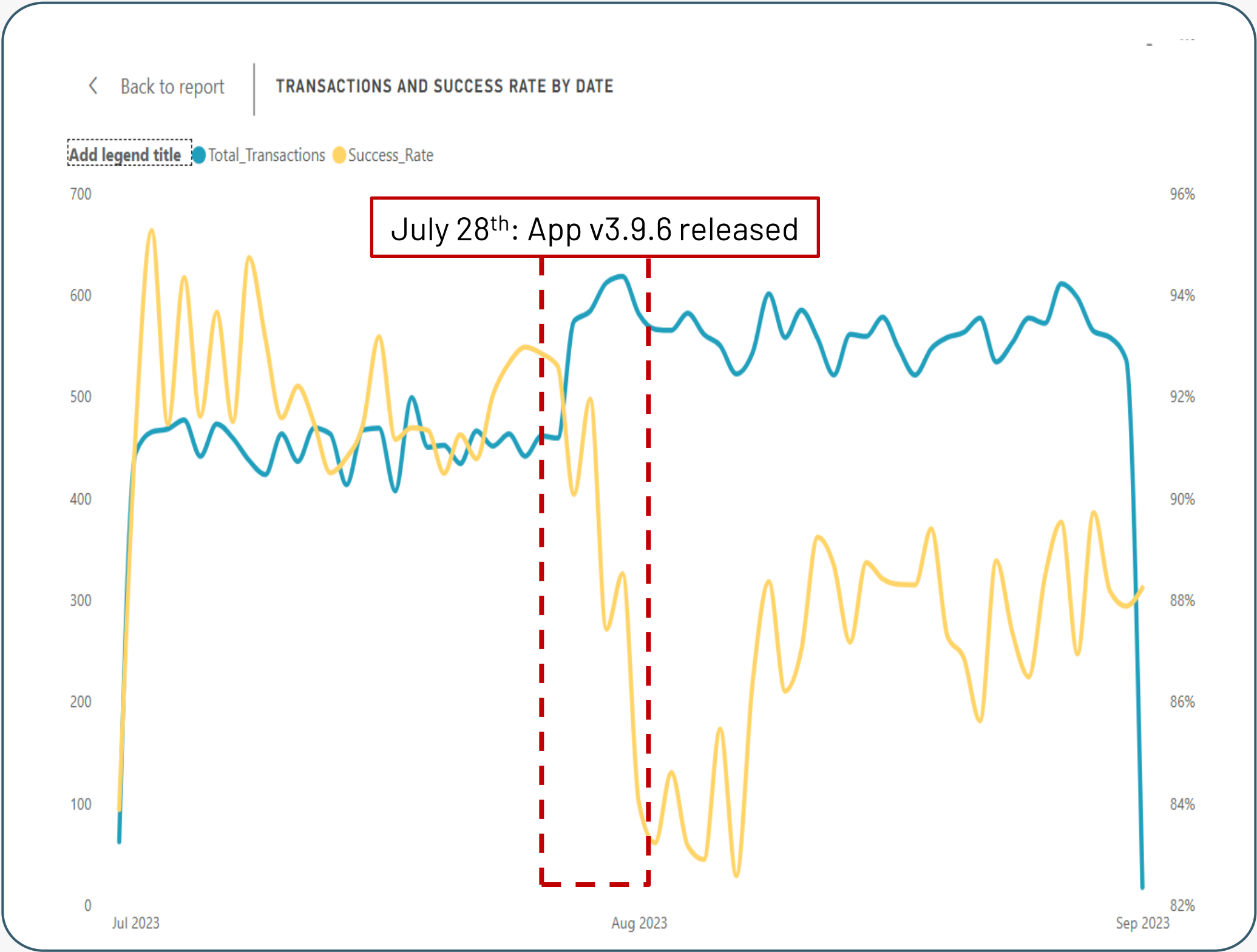
Step	User Event	Internal Stakeholder	External Stakeholders	Key Questions Addressed
01	Open and navigation	- Product Owners - UX/UI Team	-	Is the navigation flow smooth and conversion rate between each steps?
02	Checkout & Fee Review	- Product Owners - Engineering & Backend Teams	Billers / Merchants	Are promotion codes, fees, and total amounts calculated correctly at checkout?
03	Authentication & Gateway Processing	- Engineering & Backend Teams - Product Owners	Payment Gateways Issuing Banks Telecom Providers (SMS OTP)	- What causes payment failures (API timeout, processing error, deadlock) during 3DS/OTP verification? - How do gateway & bank network latencies impact transaction completion rates?
04	Receipt Confirmation & Post-transaction	- Data & BI Analysts - Customer Support - Marketing & Retention	Billers / Merchants	- What specific failure causes and error logs are reported by the system upon completion? - How fast is the transaction status updated, and is real-time support provided upon failure? What are the core metrics (CSAT/NPS, Drop-off rate) to monitor and rebuild user trust post-update?

Key Finding #1: Total success rate of all transactions declined after app update on July 28th

+20%
Total Transactions*

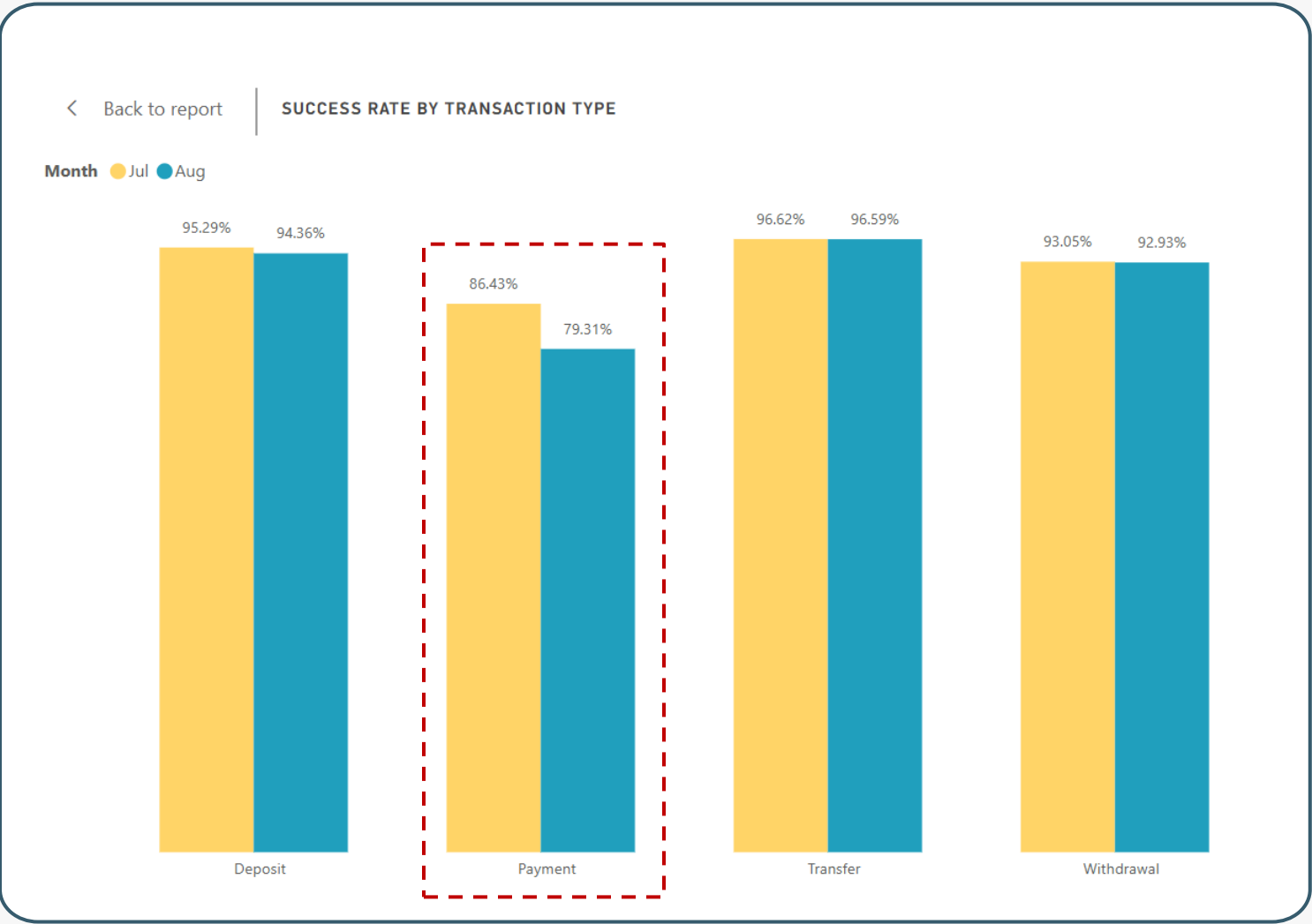
92% → 87%
Payment Success Rate*

* Between July and August 2023



Key Finding #2: Billing payment held the greatest share, but saw the largest success rate drop

Payment success rate saw the greatest decrease of **+7%**, indicating technical or compliance issues related to billers/providers.

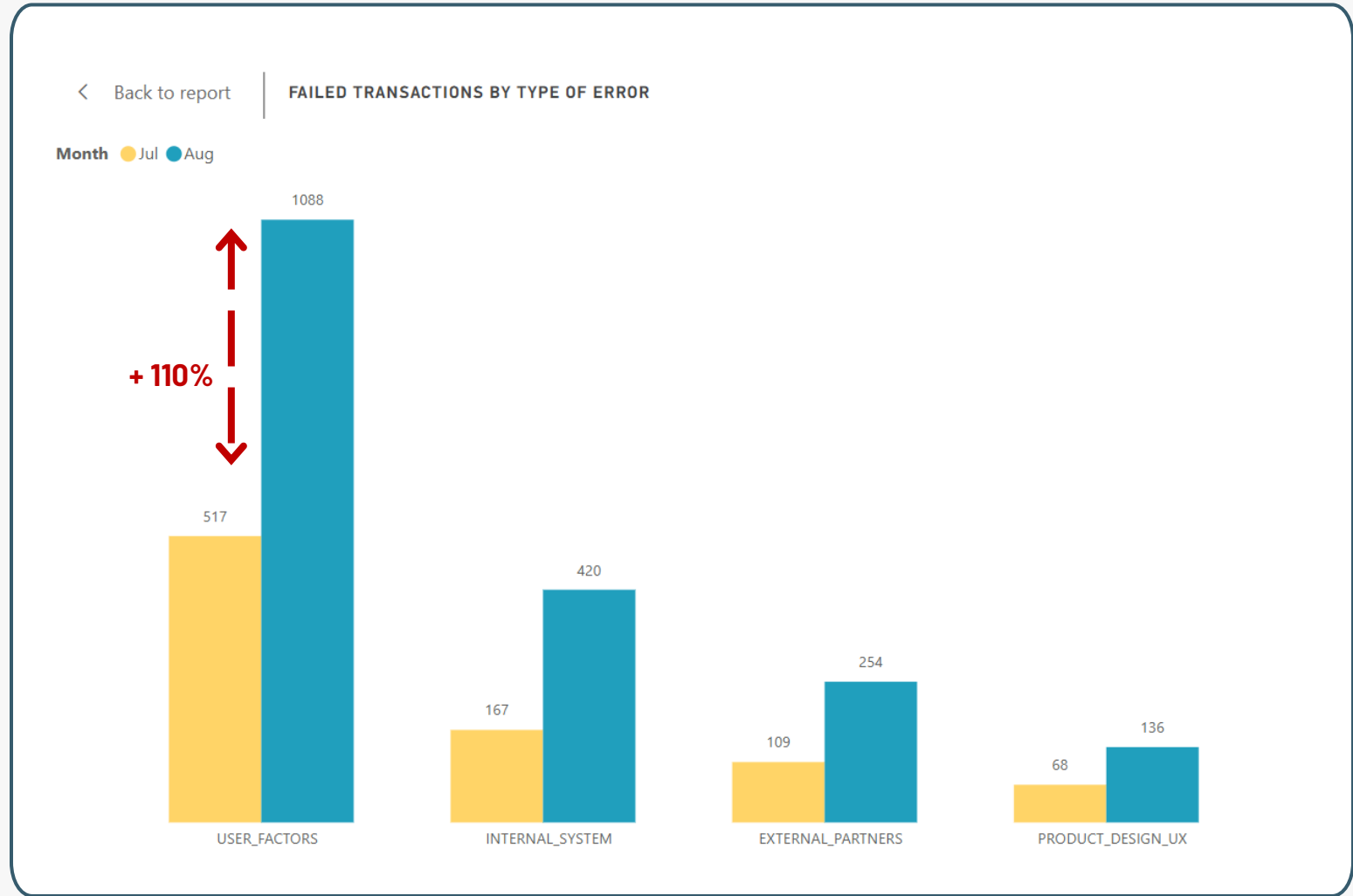


Billing holds the largest share (**37%**) but dropped by **14%** in success rate (from 86% to 72%), leading to about \$293K in revenue lost.

Category	SR_7	% Change	% Share	SR_8
Billing	86%	-14%	37%	72%
FnB	86%	-4%	24%	82%
Marketplace	87%	-2%	15%	85%
Movies	85%	-6%	11%	79%
Shopping	87%	-1%	7%	86%
Telco	84%	-5%	3%	79%
Transportation	86%	-2%	3%	84%
Traveling	88%	-7%	2%	81%

#1: Authentication and Backend Failures were the main error drivers

- Errors from User Factors surged by **+110%** from July to August.
- **Authentication issues were the main contributor (44%)** among user factors, while **backend failures drove internal system errors (8.17%)**.

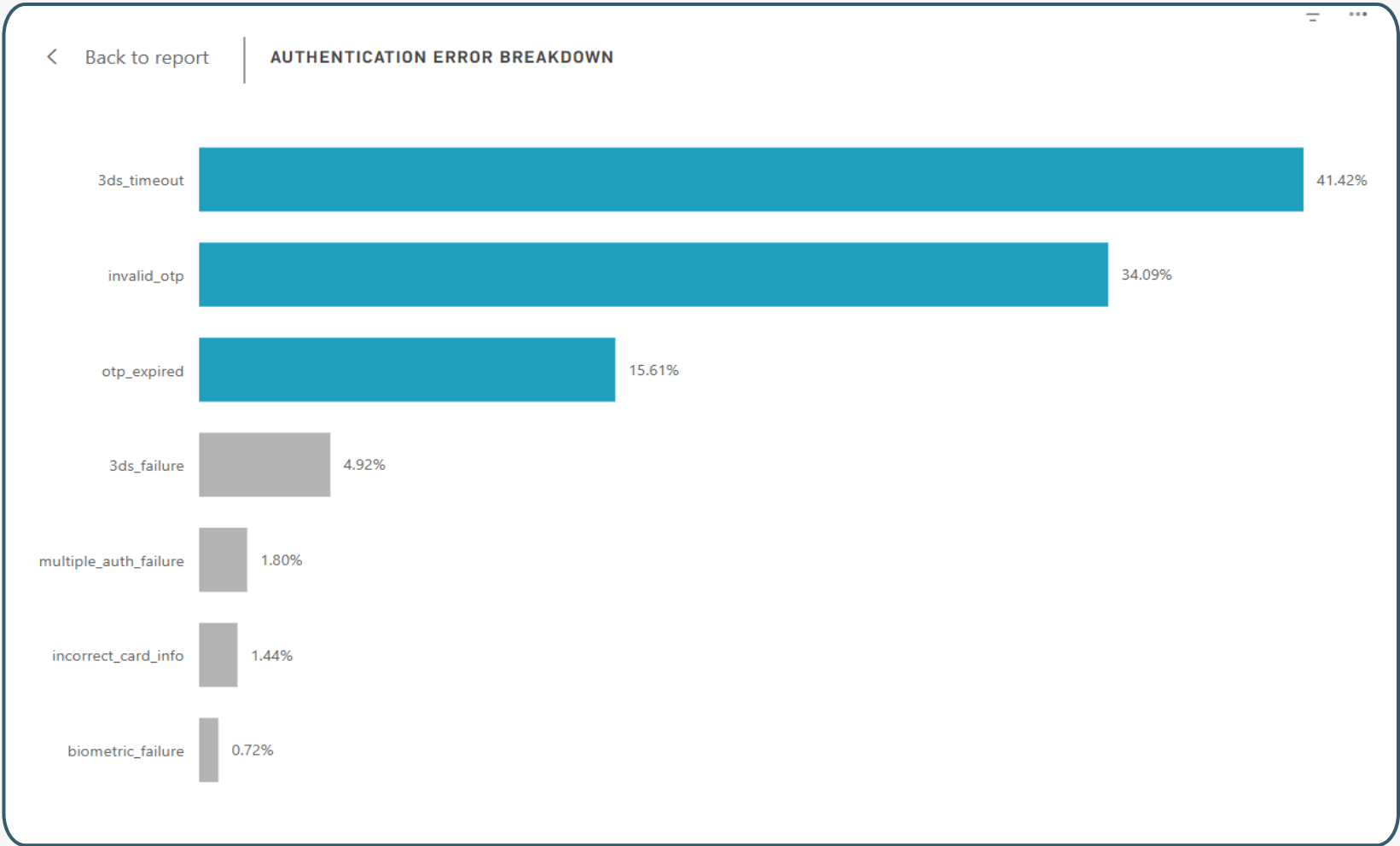


Back to report | ERROR REASON BREAKDOWN

Error Type	Total_Transactions	%GT Total_Transactions
USER_FACTORS	1,609.00	58.15%
AUTHENTICATION_ISSUES	1,199.00	43.33%
USER_BEHAVIOR	161.00	5.82%
FINANCIAL_STATUS	150.00	5.42%
DEVICE_CONNECTIVITY	99.00	3.58%
INTERNAL_SYSTEM	588.00	21.25%
BACKEND_FAILURES	226.00	8.17%
SECURITY_RISK	188.00	6.79%
INFRASTRUCTURE	174.00	6.29%
EXTERNAL_PARTNERS	365.00	13.19%
ISSUING_BANKS	153.00	5.53%
PAYMENT_GATEWAYS	150.00	5.42%
INTEGRATION_ISSUES	62.00	2.24%
PRODUCT_DESIGN_UX	205.00	7.41%
FLOW_COMPLEXITY	101.00	3.65%
TIMING_SESSION	63.00	2.28%
VISUAL_FEEDBACK	41.00	1.48%
Total	2,767.00	100.00%

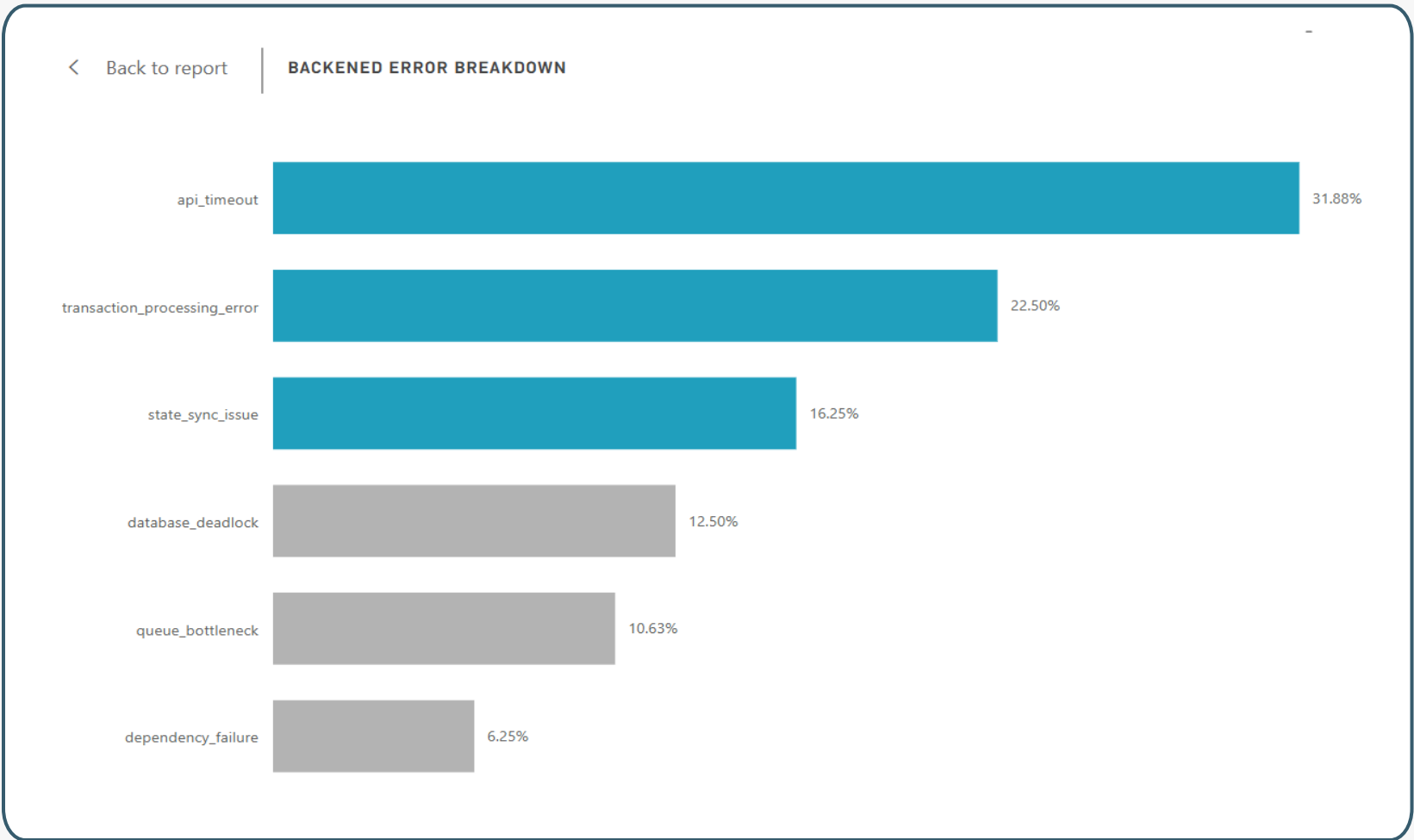
#1: Authentication and Backend Failures were the main drivers

- **Authentication Errors:** Dominated by time-related issues, led by **3DS timeout (41.4%)**, **invalid OTP (34.1%)**, and **OTP expired (15.6%)**.
- **Backend Errors:** Driven primarily by **API timeout (31.9%)** and transaction processing errors.



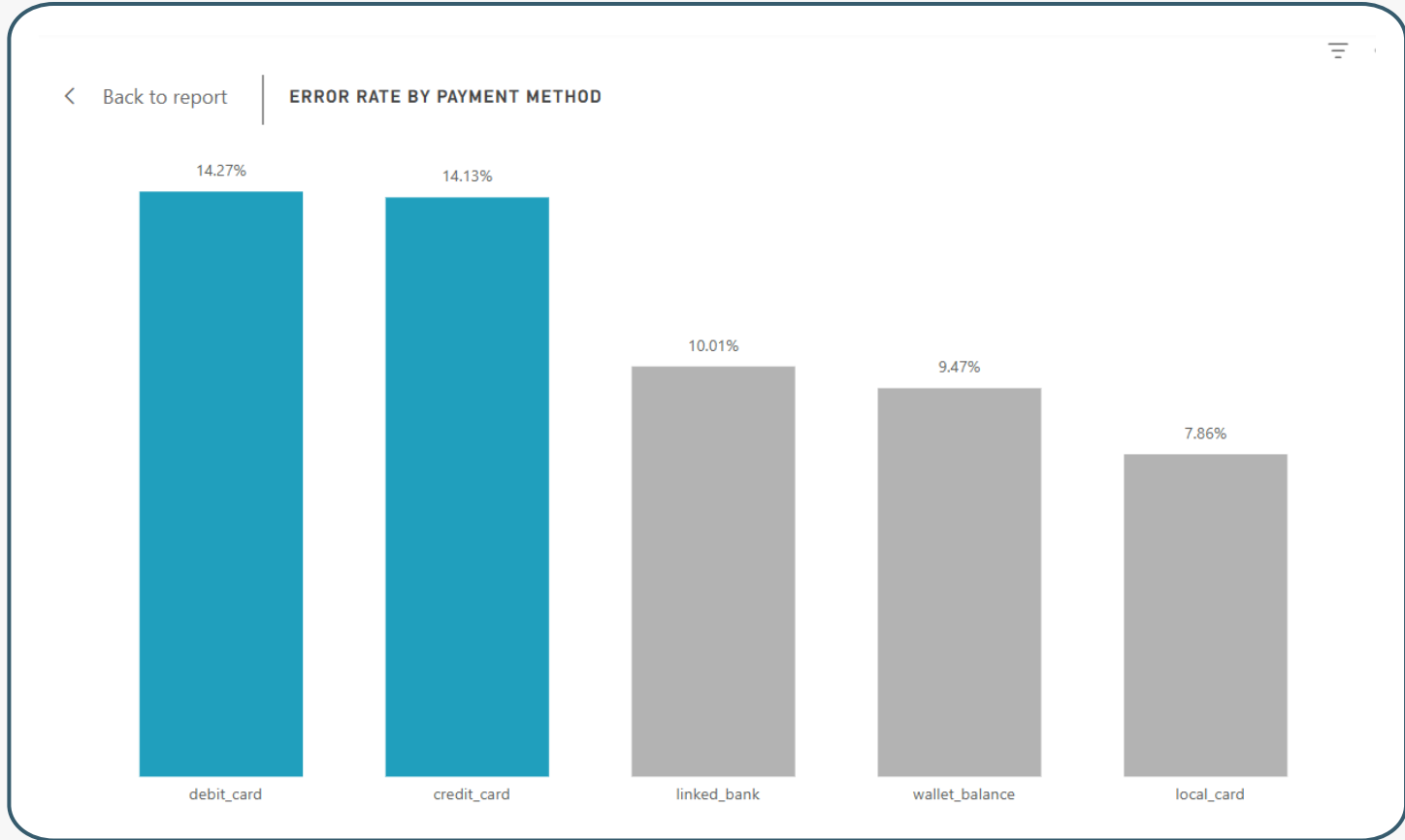
Hypotheses:

- **Session & Timeout Policy:** App v3.9.6 shortened session duration to 15 mins, causing OTP expiration and 3DS latency before user completion.
- **Backend Bottleneck:** Multiple retry requests overloaded backend queues, causing API timeouts (31.9%) and OTP sync failures.
- **UI/UX Friction:** Lack of visible countdown timers to make users aware of time limits.

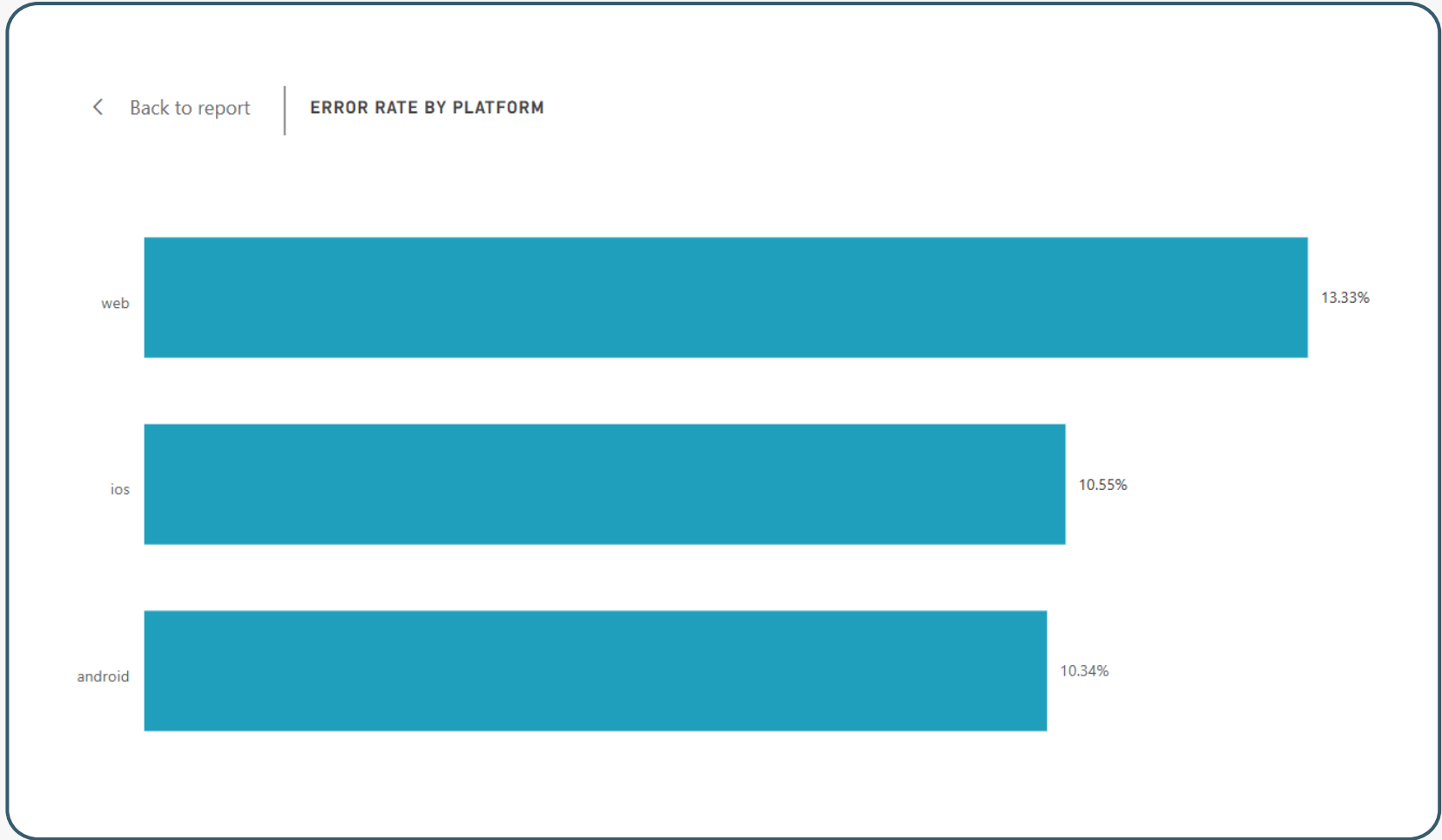


#2: Credit Cards and Web Platform drove the highest error rates

- Credit & Debit Cards recorded the highest error rate at approximately **14%**, despite lower overall transaction volume.
- This highlights friction in third-party banking flows, confirming the need to improve 3DS and OTP authentication.

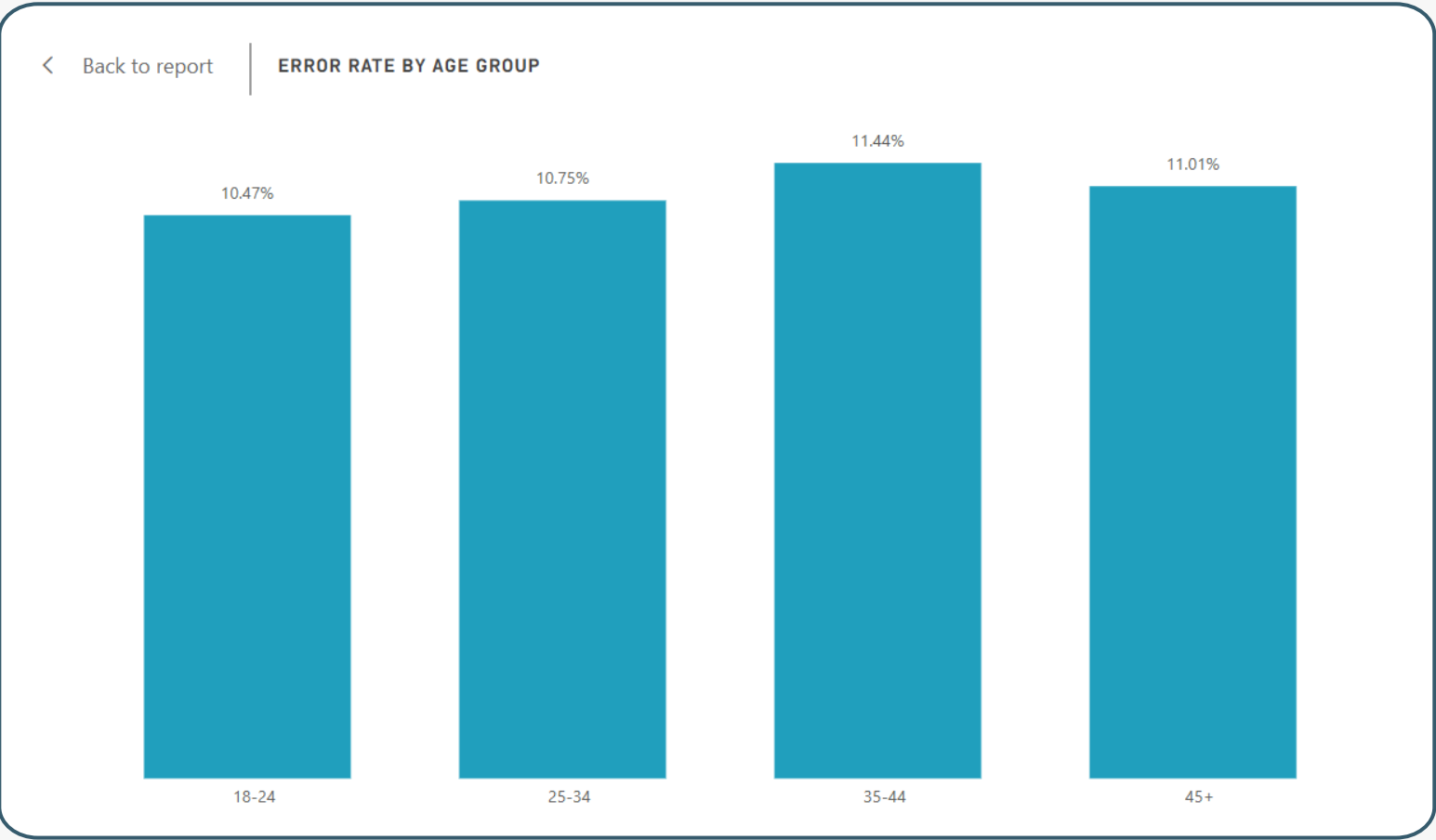


- Web-based transactions recorded the **highest failure rate at 13.33%**.
- Mobile error rates are slightly lower and highly comparable between iOS (10.55%) and Android (10.34%).
- Key Takeaway: While Web suffers from a higher error rate, mobile apps still account for the majority of absolute failure volume due to higher transaction traffic.



#3: 35–44 age group and inactive users presented the highest risk

Overall, there is not much difference in error rate across different age groups. However, 35-44 recorded the highest error rate at **11.44%**.



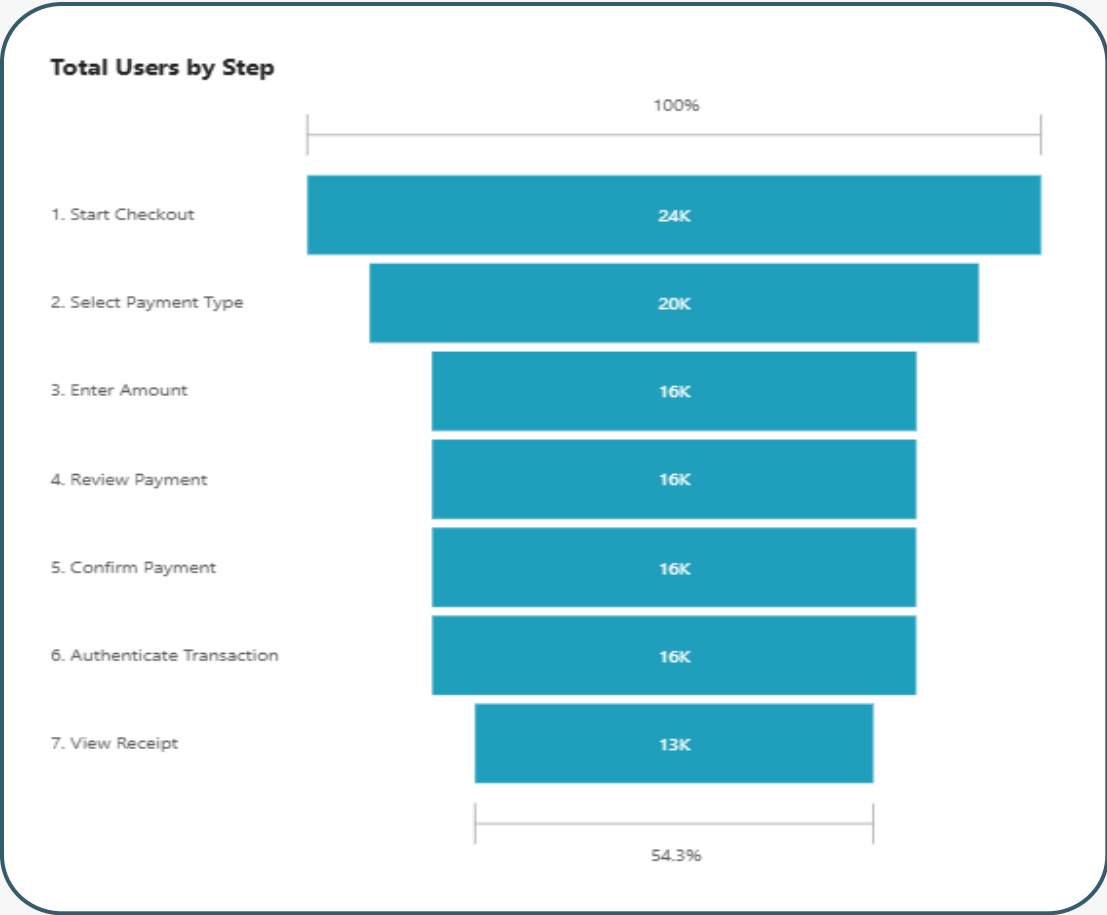
- Inactive users (**Dormant & Low activity**) face the **highest error rate (14%)** and spend the most time per session.
- Key Takeaway: Long-term inactive users struggle with checkout navigation—indicating UX/UI friction that requires clearer guidance and real-time support.

Back to report ACTIVITY LEVEL OVERVIEW			
Activity Level	Account Age (days)	Session Duration (secs)	Error Rate
dormant	255.40	89.74	14.06%
low	249.70	85.29	12.00%
high	243.10	82.64	9.82%
medium	234.50	83.35	10.41%
Total	245.23	84.44	10.90%

**Dormant: Users who used to use the service but stopped*

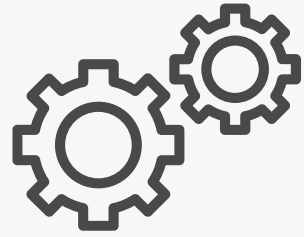
#4: The post-authentication phase recorded the most drop-offs

- Overall conversion rate was **54.3%**, with 13K users completing the receipt stage.
- Immediate friction occurs when **selecting the payment type**, where 17% of users (4K) exit instantly.
- A major bottleneck follows authentication with an 18% drop-off; resolving this will directly boost overall success rates.
- Total time-to-convert averages **141.5 seconds (~2.5 mins)**.
- Users spend only ~5 seconds on Step 2, suggesting drop-offs stem from misclicks rather than UX complexity. → Improve UX/UI to avoid confusion.
- **Authentication duration is the longest (31.6s)**. *Note: Version 3.9.9 addressed this by expanding gateway connectivity and increasing transaction timeouts.*



Step	Total Users	%First Step	%Previous Step	Avg time from First Step	Avg Time from Previous Step
1. Start Checkout	23,551	100%		0.00	
2. Select Payment Type	19,551	83%	83%	5.49	5.49
3. Enter Amount	15,551	66%	80%	28.80	23.28
4. Review Payment	15,541	66%	100%	42.69	13.98
5. Confirm Payment	15,532	66%	100%	71.61	28.79
6. Authenticate Transaction	15,531	66%	100%	97.12	31.63
7. View Receipt	12,784	54%	82%	141.53	27.68

Recommendations



Reduce authentication & backend friction

- **Authentication optimization:** Resolve 3DS and OTP timeouts by identifying root causes: whether session limits are too short or UX/UI guidance lacks time-limit notifications.
- **Backend infrastructure:** Review the v3.9.9 deployment to fix payment gateway API bugs and resolve connectivity bottlenecks.
- **Partner collaboration:** Conduct card-specific debugging with billers and acquiring banks to address Credit/Debit failures.



Payment flow & UX optimization

- **A/B Testing:** Test modified authentication flows (specifically post-v3.9.8 redesign) to identify optimal conversion steps.
- **UX Research:** Survey users during checkout and authentication to eliminate friction when selecting payment methods.
- **Alternative method:** Promote high-success payment methods (e.g., Wallet Balance) while Card Payment issues are being resolved.



Real-time support for high-risk users

- **Targeted assistance:** Apply AI Chatbot support and in-app real-time guidance specifically tailored for Dormant and Low-activity users.
- **In-time support:** Provide instant, contextual troubleshooting messages when errors occur during authentication.
- **Re-engagement campaigns:** Retarget dropped-off users via Email and Push Notifications to educate them on payment flow updates.